Nonmonotonic Preferences

There are $I = 2$ consumers and $L = 2$ goods and a single firm whose only ability is free disposal. The endowments are $\boldsymbol{\omega}_1 = (0, \bar{\omega}_2)$ and $\boldsymbol{\omega}_2 = (\bar{\omega}_1, 0)$ and each consumer owns an equal share in the firm. Each consumer's utility function is given by $u_1(x_{11}, x_{21}) = \sqrt{x_{11}} + x_{21}$ and $u_2(x_{12}, x_{22}) = x_{11}$. Notice that consumer 2's preferences are nonmonotone.

We note that consumer 1's indifference curve has an infinite slope at $\boldsymbol{\omega}_1$. To find the indifference curve, we solve the utility function for good 2 as a function of the consumption of good 1 and a fixed level of utility $\bar{u}_1$:

$$\bar{u}_1 = \sqrt{x_{11}} + x_{21} \qquad \Longrightarrow \qquad x_{21}(\bar{u}_1, x_{11}) = \bar{u}_1 - \sqrt{x_{11}}$$

The slope of this is:

$$\frac{\partial x_{21}(\bar{u}_1, x_{11})}{\partial x_{11}} = -\frac{1}{2}(x_{11})^{-\frac{1}{2}}$$

At $\boldsymbol{x}_1 = \boldsymbol{\omega}_1$, this is infinite because $\omega_{11} = 0$.

Demand

We now solve for each consumer's optimal demand given prices $\boldsymbol{p} = (p_1, p_2)$.

Consumer 1's problem is:

$$\max_{\boldsymbol{x_1} \geq \boldsymbol{0}} \sqrt{x_{11}} + x_{21} \text{ subject to } p_1 x_{11} + p_2 x_{21} \leq p_2 \bar{\omega}_2$$

The budget constraint will bind at the optimum as consumer 1's utility is monotonic. So we can write the budget constraint as $x_{21} = \frac{1}{p_2}[p_2\bar{\omega}_2 - p_1 x_{11}] = \bar{\omega}_2 - \frac{p_1}{p_2}x_{11}$ and substitute it in:

$$\max_{x_{11} \geq 0} \sqrt{x_{11}} + \bar{\omega}_2 - \frac{p_1}{p_2}x_{11} \quad \text{subject to} \quad \bar{\omega}_2 - \frac{p_1}{p_2}x_{11} \geq 0$$

Notice that the nonnegativity constraint for x_{21} remains. Notice also that unless $p_2 = 0$, the consumer will always choose a positive amount of good 1. Taking first-order conditions for the interior solution case:

$$\frac{1}{2\sqrt{x_{11}}} - \frac{p_1}{p_2} = 0 \qquad \Longrightarrow \qquad x_{11} = \left(\frac{p_2}{2p_1}\right)^2$$

This bundle will only satisfy the nonnegativity constraint for good 2 (in other words, be affordable) if $\bar{\omega}_2 \geq \frac{p_2}{4p_1}$. If this binds, then the consumer spends all wealth on good 1.

Therefore:

$$x_{11}\left(\boldsymbol{p}, \boldsymbol{p} \cdot \boldsymbol{\omega}_1\right) = \max\left\{\left(\frac{p_2}{2p_1}\right)^2, \frac{p_2\bar{\omega}_2}{p_1}\right\}$$

For good 2 in the case where $p_2 > 0$, the consumer will demand either zero (if $\bar{\omega}_2 \leq \frac{p_2}{4p_1}$) or they spend all their remaining income after purchasing $\frac{p_2^2}{4p_1^2}$ units of good 1. In this case, the total amount spent on good 1 is $\frac{p_2^2}{4p_1}$. Therefore they have $p_2\bar{\omega}_2 - \frac{p_2^2}{4p_1}$ left to spend, so they purchase:

$$\frac{1}{p_2}\left[p_2\bar{\omega}_2 - \frac{p_2^2}{4p_1}\right] = \bar{\omega}_2 - \frac{p_2}{4p_1}$$

For good 2 in the case where $p_2 = 0$, they simply demand an infinite amount of it. So the consumer's demand for good 2 is:

$$x_{21}\left(\boldsymbol{p}, \boldsymbol{p} \cdot \boldsymbol{\omega}_1\right) = \begin{cases} \max\left\{0, \bar{\omega}_2 - \frac{p_2}{4p_1}\right\} & \text{if } p_2 > 0 \\ \infty & \text{if } p_2 = 0 \end{cases}$$

Consumer 2's problem is much simpler. If the price of good 1 is positive, they will spend their entire endowment on good 1. If the price of good 1 is zero, they will demand an infinite amount of it:

$$\boldsymbol{x}_2\left(\boldsymbol{p}, \boldsymbol{p} \cdot \bar{\boldsymbol{\omega}}_2\right) = \begin{cases} (\bar{\omega}_1, 0) & \text{if } p_1 > 0 \\ (\infty, 0) & \text{if } p_1 = 0 \end{cases}$$

Equilibrium

We need to find a vector of prices $\boldsymbol{p}$ such that the market clears in both goods. However, we will find that finding such a price vector is impossible.

Consider each possible case:

- $p_1 > 0$ and $p_2 > 0$. Consumer 1 is a net demander of good 1 (demands a positive amount but has zero endowment) but consumer 2 demands only the endowment. Therefore this can't be an equilibrium.

- $p_1 = 0$ and $p_2 > 0$. Both consumers demand an infinite amount of good 1. Therefore this can't be an equilibrium.

- $p_1 > 0$ and $p_2 = 0$. Consumer 1 demands an infinite amount of good 2. Therefore this can't be an equilibrium.